EXPERIMENTS OF 25 TO 28

25.

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>
#include <dirent.h>

int main() {
    int fd;
    char buffer[100];
    ssize_t n;
    struct stat fileInfo;
    struct dirent *entry;
    DIR *dir;

    /* Create and open a file */
    fd = open("sample.txt", O_CREAT | O_RDWR | O_TRUNC,
0644);

    if (fd == -1) {
        perror("Error opening file");
        return 1;
    }
```

```
/* Write data */
if (write(fd, "UNIX I/O System Calls", 21) == -1) {
    perror("Write error");
    close(fd);
    return 1;
}
printf("File created and data written.\n");

/* fcntl system call */
if (fcntl(fd, F_GETFL) == -1) {
    perror("fcntl error");
    close(fd);
    return 1;
}
printf("fcntl executed successfully.\n");

/* lseek system call */
if (lseek(fd, 0, SEEK_SET) == -1) {
    perror("lseek error");
    close(fd);
    return 1;
}

n = read(fd, buffer, sizeof(buffer) - 1);
if (n == -1) {
```

```
    perror("Read error");
    close(fd);
    return 1;
}
buffer[n] = '\0';
printf("File contents: %s\n", buffer);
close(fd);

/* stat system call */
if (stat("sample.txt", &fileInfo) == -1) {
    perror("stat error");
    return 1;
}
printf("File size: %ld bytes\n", (long)fileInfo.st_size);

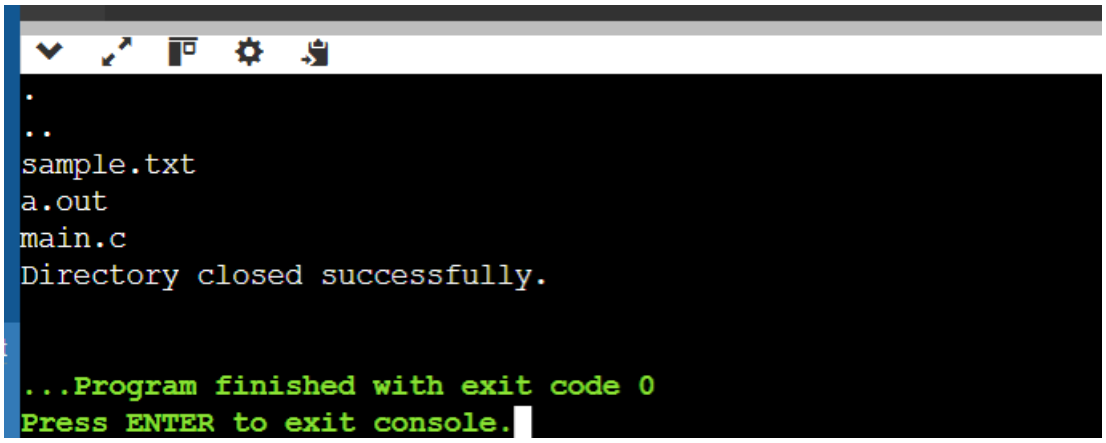
/* opendir and readdir system calls */
dir = opendir(".");
if (dir == NULL) {
    perror("opendir error");
    return 1;
}

printf("Files in current directory:\n");
while ((entry = readdir(dir)) != NULL) {
    printf("%s\n", entry->d_name);
}
```

```
    closedir(dir);
    printf("Directory closed successfully.\n");

    return 0;
}
```

OUTPUT:



26.

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    FILE *fp;
    char data[200];
    int choice;

    while (1) {
        printf("\n--- File Management Operations ---\n");
        printf("1. Create and Write File\n");
        printf("2. Read File\n");
        printf("3. Append to File\n");
        printf("4. Delete File\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice) {
            case 1:
                fp = fopen("sample.txt", "w");
                if (fp == NULL) {
                    printf("Error creating file.\n");
```

```
            break;
        }

        printf("Enter text to write: ");
        fgets(data, sizeof(data), stdin);
        fputs(data, fp);
        fclose(fp);
        printf("File created and written successfully.\n");
        break;

    case 2:
        fp = fopen("sample.txt", "r");
        if (fp == NULL) {
            printf("File does not exist.\n");
            break;
        }

        printf("File contents:\n");
        while (fgets(data, sizeof(data), fp) != NULL) {
            printf("%s", data);
        }
        fclose(fp);
        printf("\n");
        break;

    case 3:
```

```
    fp = fopen("sample.txt", "a");
    if (fp == NULL) {
        printf("Error opening file.\n");
        break;
    }

    printf("Enter text to append: ");
    fgets(data, sizeof(data), stdin);
    fputs(data, fp);
    fclose(fp);
    printf("Data appended successfully.\n");
    break;

case 4:
    if (remove("sample.txt") == 0)
        printf("File deleted successfully.\n");
    else
        printf("Unable to delete file.\n");
    break;

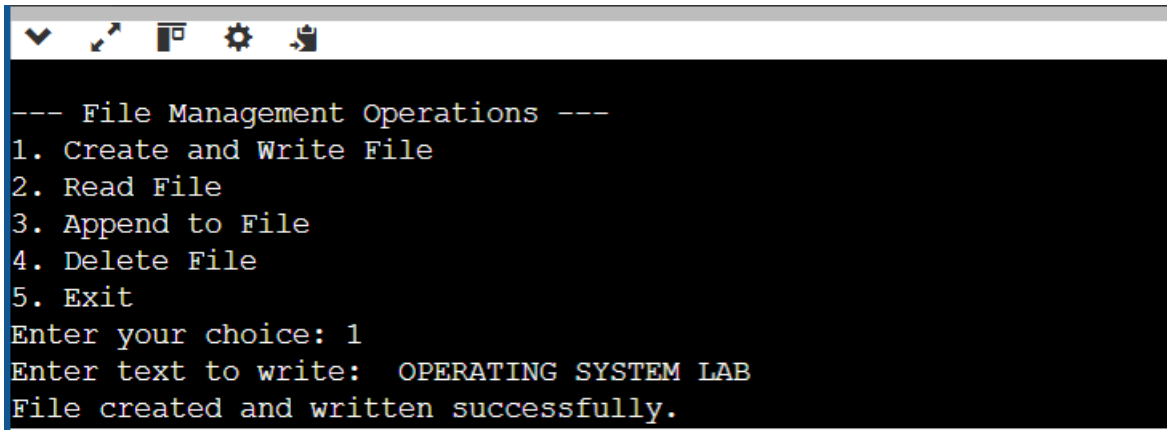
case 5:
    printf("Exiting program.\n");
    return 0;

default:
    printf("Invalid choice. Try again.\n");
```

```
        }
    }

    return 0;
}
```

OUTPUT:



27.

```
#include <stdio.h>
#include <dirent.h>

int main() {
    DIR *dir;
    struct dirent *entry;

    dir = opendir(".");

    if (dir == NULL) {
        perror("Unable to open directory");
        return 1;
    }

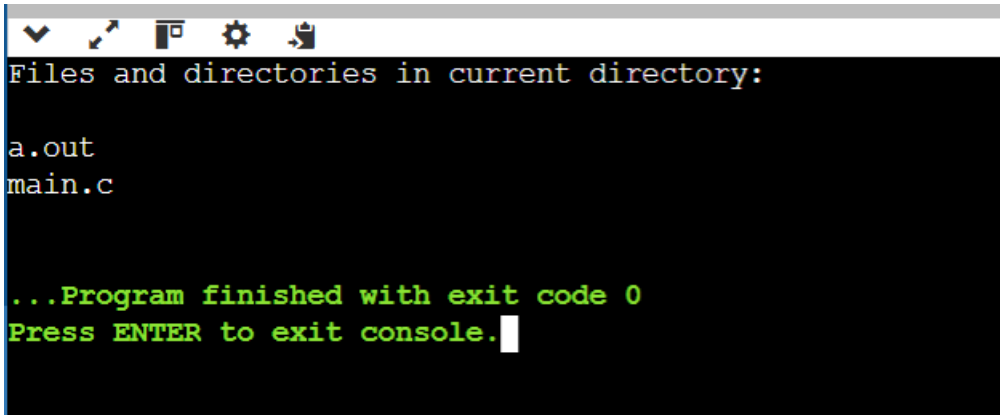
    printf("Files and directories in current directory:\n\n");

    while ((entry = readdir(dir)) != NULL) {
        if (entry->d_name[0] != '.') {
            printf("%s\n", entry->d_name);
        }
    }

    closedir(dir);
```

```
    return 0;
}
```

OUTPUT:



28.

```
#include <stdio.h>
#include <string.h>

int main() {
    FILE *fp;
    char filename[100];
    char pattern[100];
    char line[500];
    int found = 0;

    printf("Enter file name: ");
    scanf("%99s", filename);

    printf("Enter word to search: ");
    scanf("%99s", pattern);

    fp = fopen(filename, "r");

    if (fp == NULL) {
        printf("Error: File not found.\n");
        return 1;
    }

    printf("\nMatching lines:\n");
```

```
    while (fgets(line, sizeof(line), fp) != NULL) {
        if (strstr(line, pattern) != NULL) {
            printf("%s", line);
            found = 1;
        }
    }

    if (!found) {
        printf("No matching lines found.\n");
    }

    fclose(fp);

    return 0;
}
```

OUTPUT:

